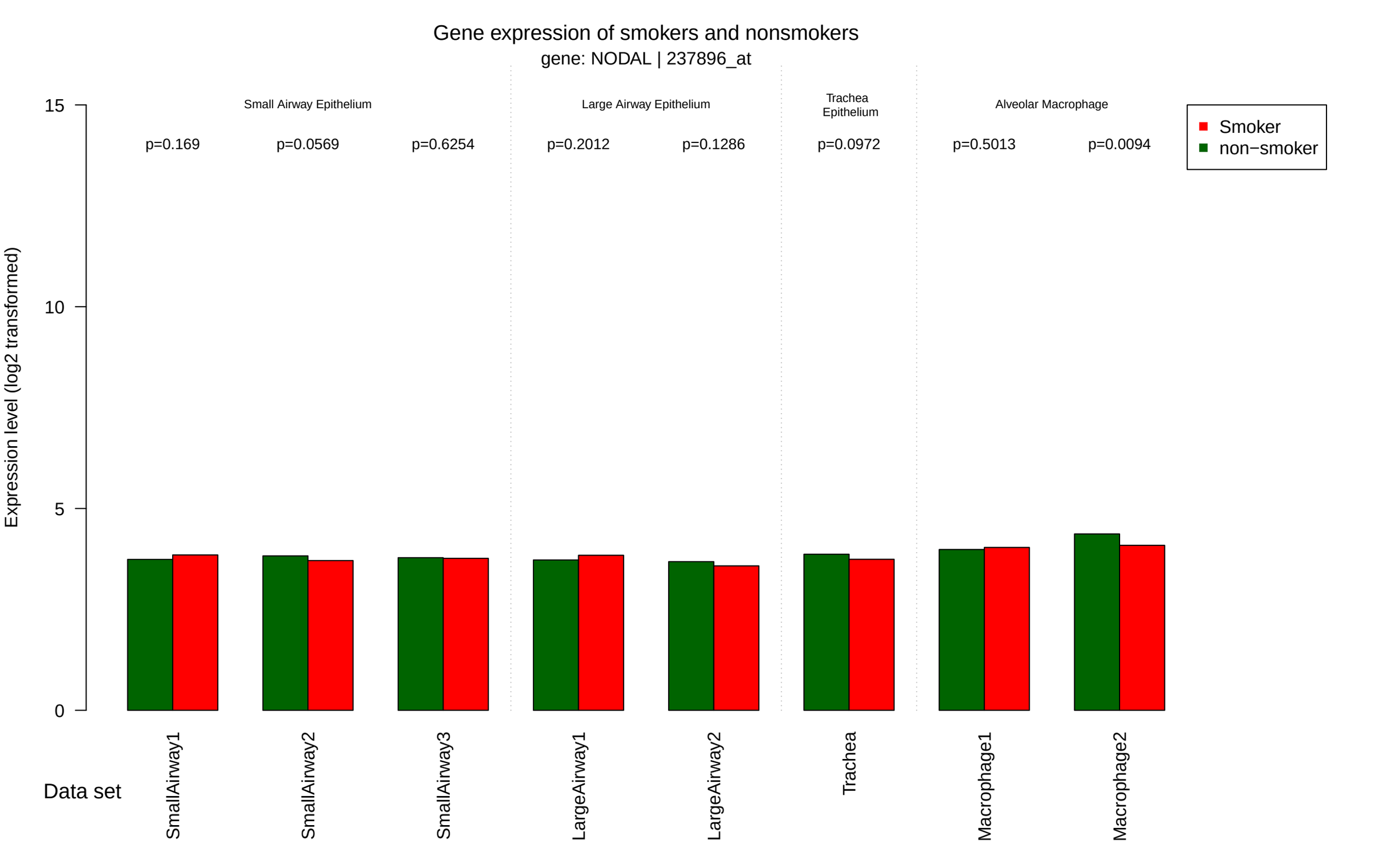
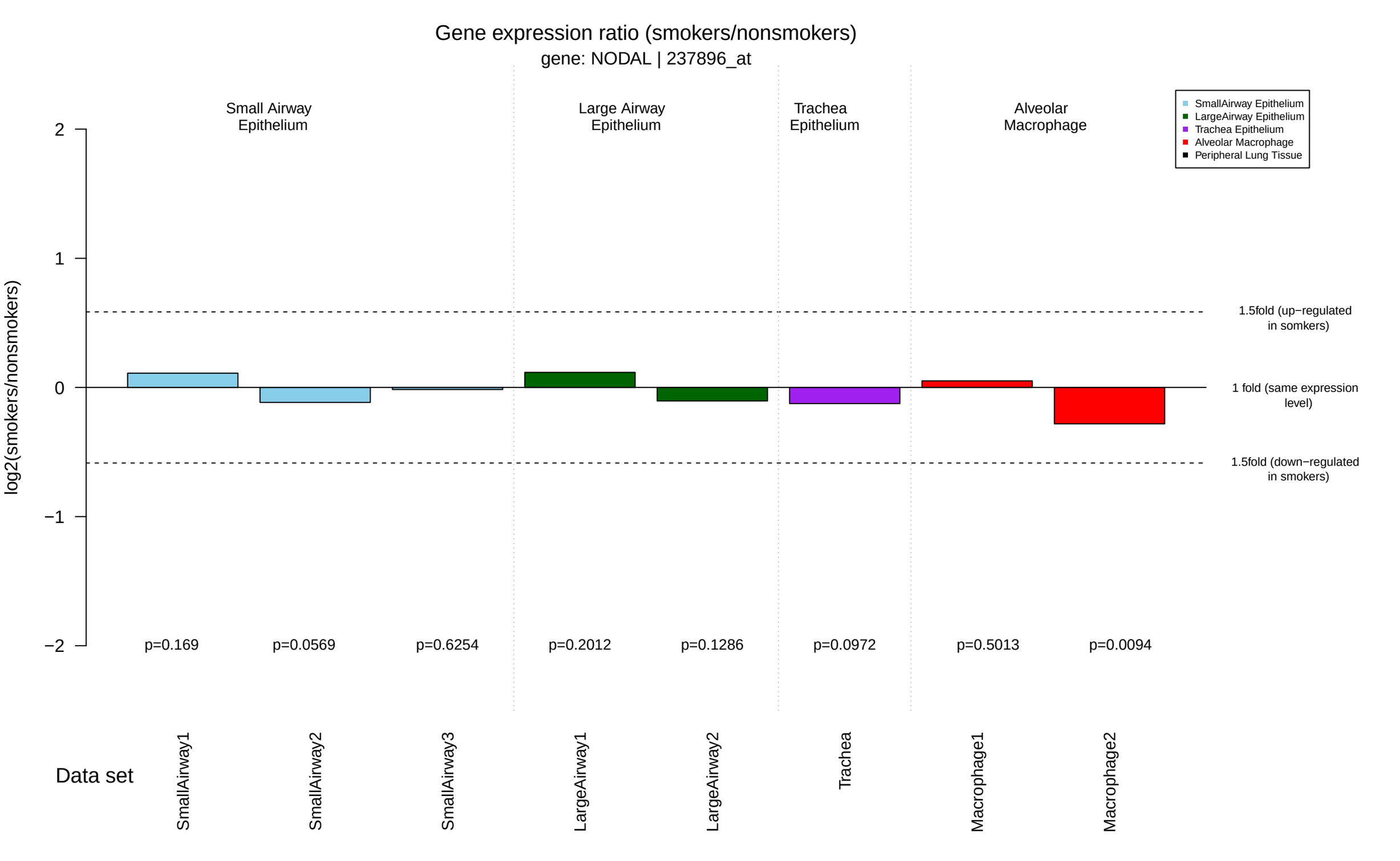
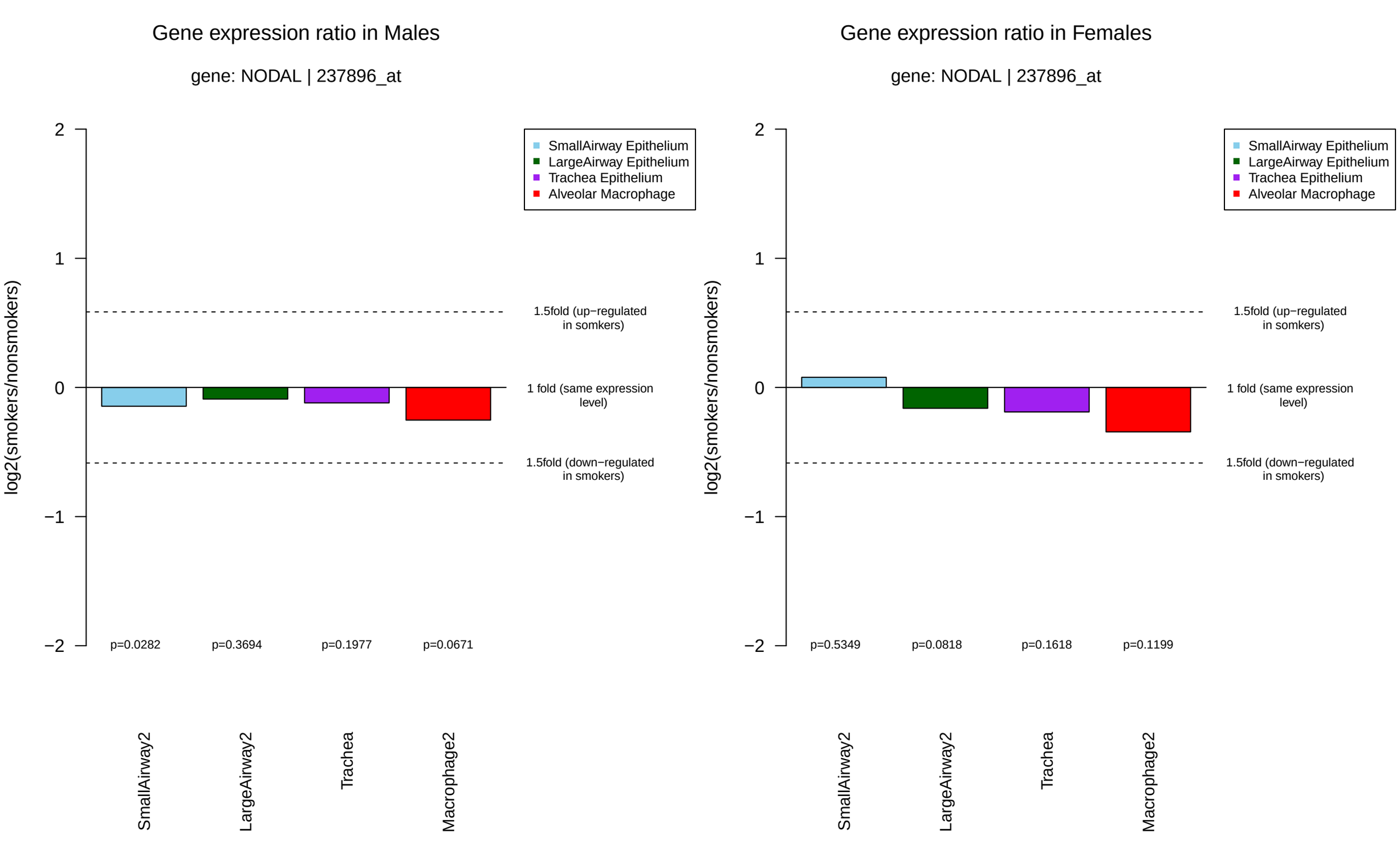
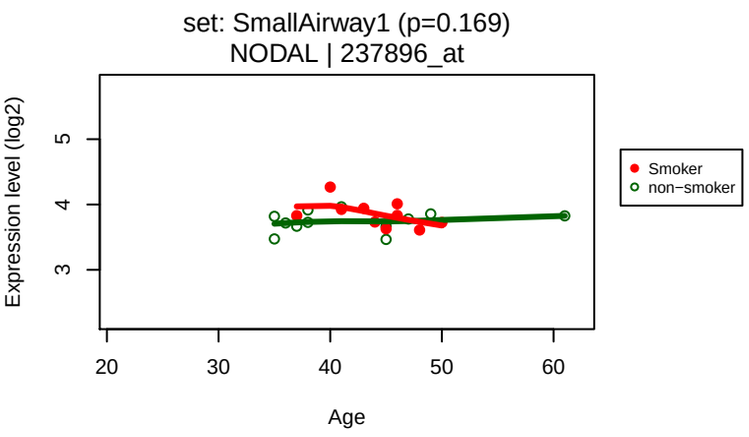
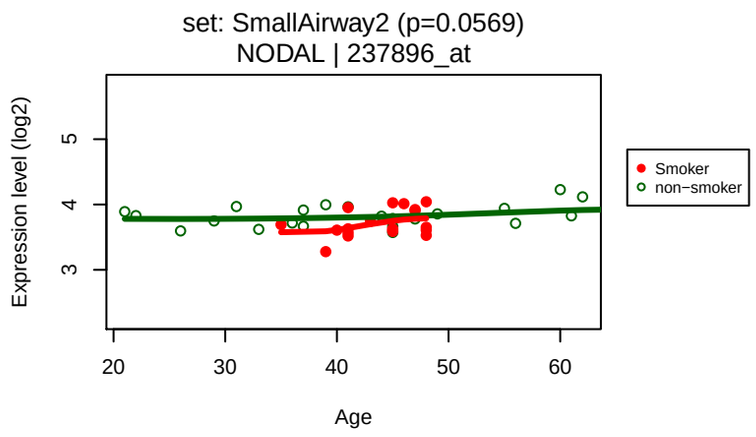
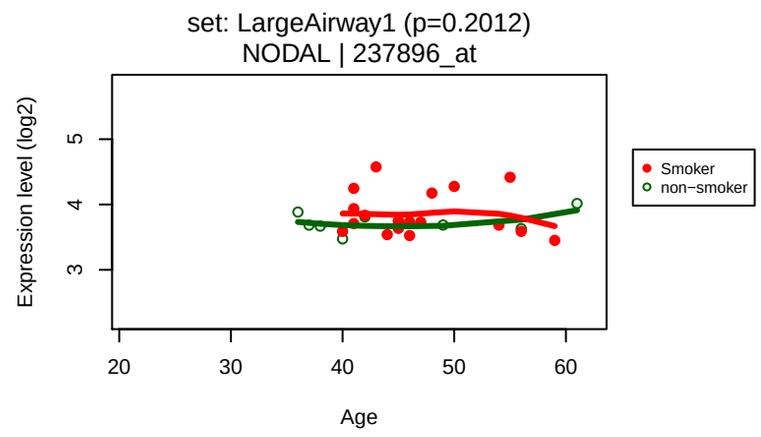

Gene expression of smokers and nonsmokers
gene: NODAL | 237896_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.169
p=0.0569
p=0.6254
p=0.2012
p=0.1286
p=0.0972
p=0.5013
p=0.0094
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

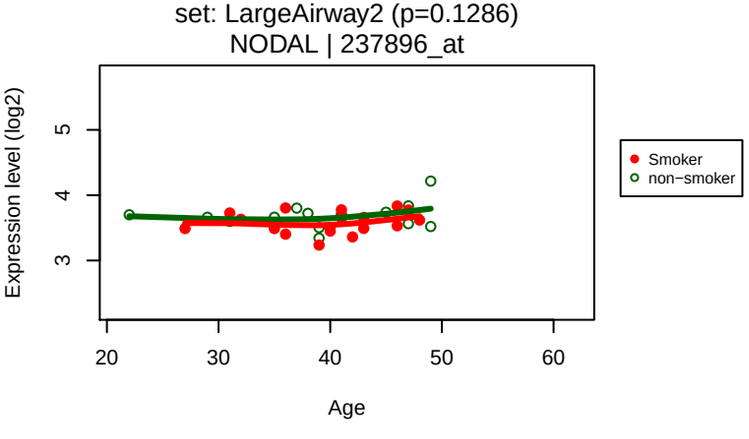

Gene expression ratio (smokers/nonsmokers)
gene: NODAL | 237896_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.169
p=0.0569
p=0.6254
p=0.2012
p=0.1286
p=0.0972
p=0.5013
p=0.0094
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

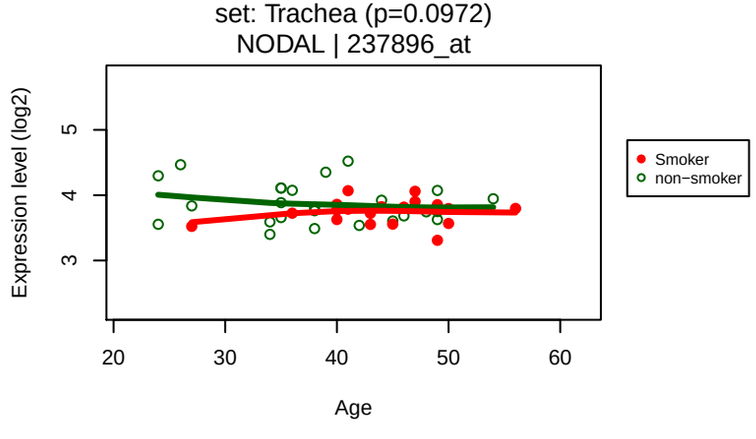

Gene expression ratio in Males
gene: NODAL | 237896_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0282
p=0.3694
p=0.1977
p=0.0671
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NODAL | 237896_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5349
p=0.0818
p=0.1618
p=0.1199
SmallAirway2
LargeAirway2
Trachea
Macrophage2

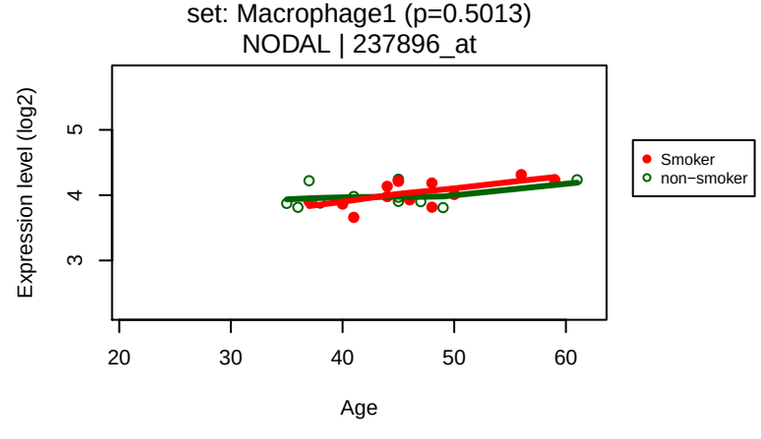

set: SmallAirway1 (p=0.169)
NODAL | 237896_at
Expression level (log2)
Age
Smoker
non−smoker

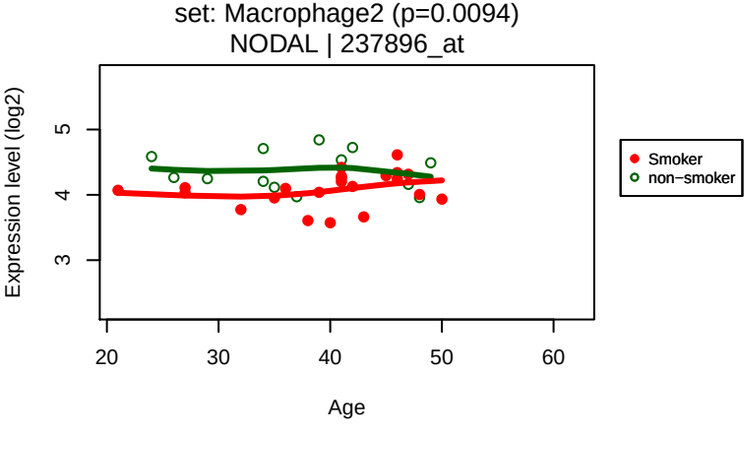

set: SmallAirway2 (p=0.0569)
NODAL | 237896_at
Expression level (log2)
Age
Smoker
non−smoker

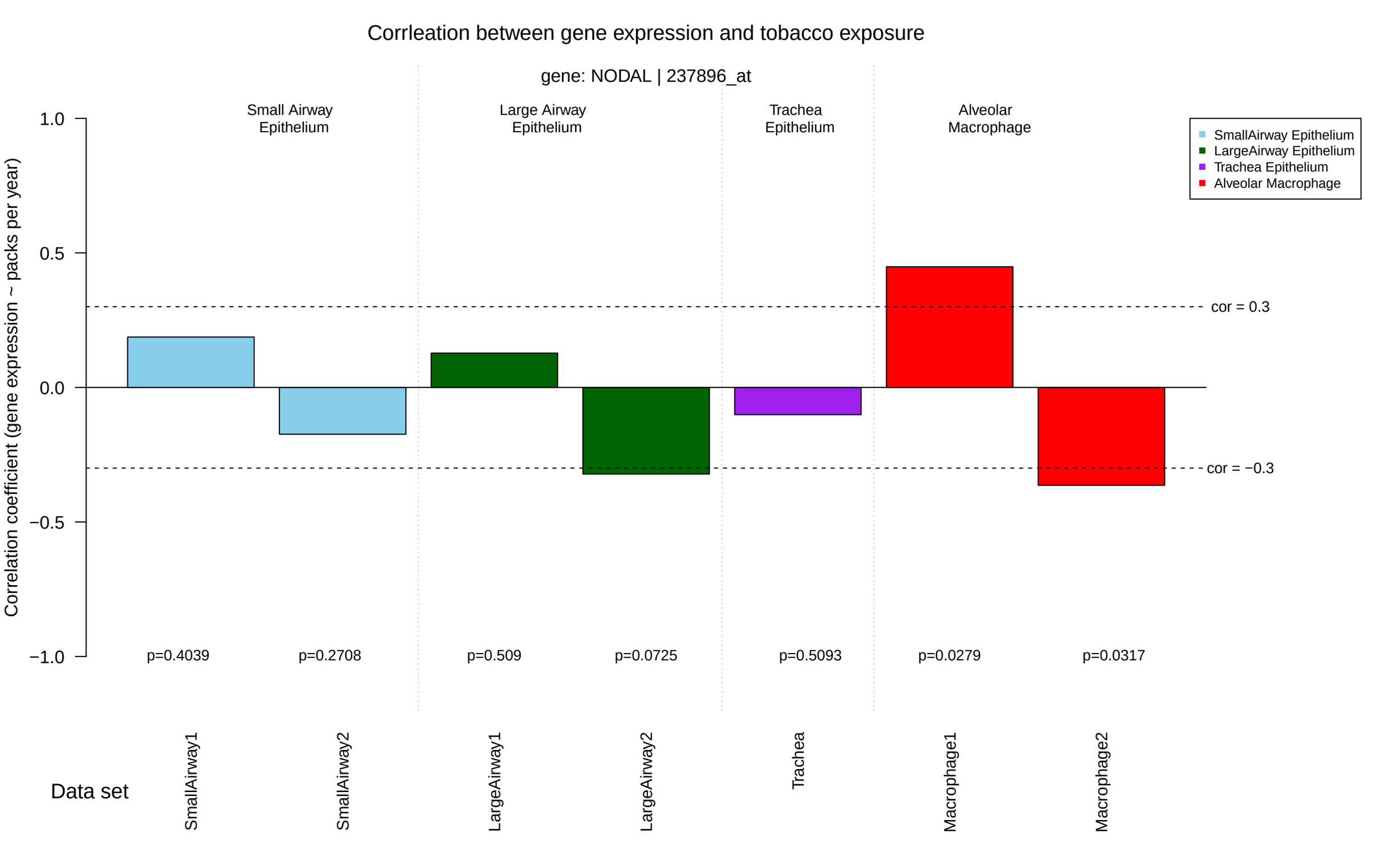

set: LargeAirway1 (p=0.2012)
NODAL | 237896_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1286)
NODAL | 237896_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0972)
NODAL | 237896_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5013)
NODAL | 237896_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0094)
NODAL | 237896_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NODAL | 237896_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4039
p=0.2708
p=0.509
p=0.0725
p=0.5093
p=0.0279
p=0.0317
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2